What does this trial test?

HKD programme

ERD programme



Weight change at 6 months

Randomisation · shared support · objective weighing

Lim et al. (2024), Nutrients 16:4380 · Methods 2.4–2.7 · doi:10.3390/nu16244380

A calorie gap emerged.

HKD

−704

kcal/day

ERD

−423

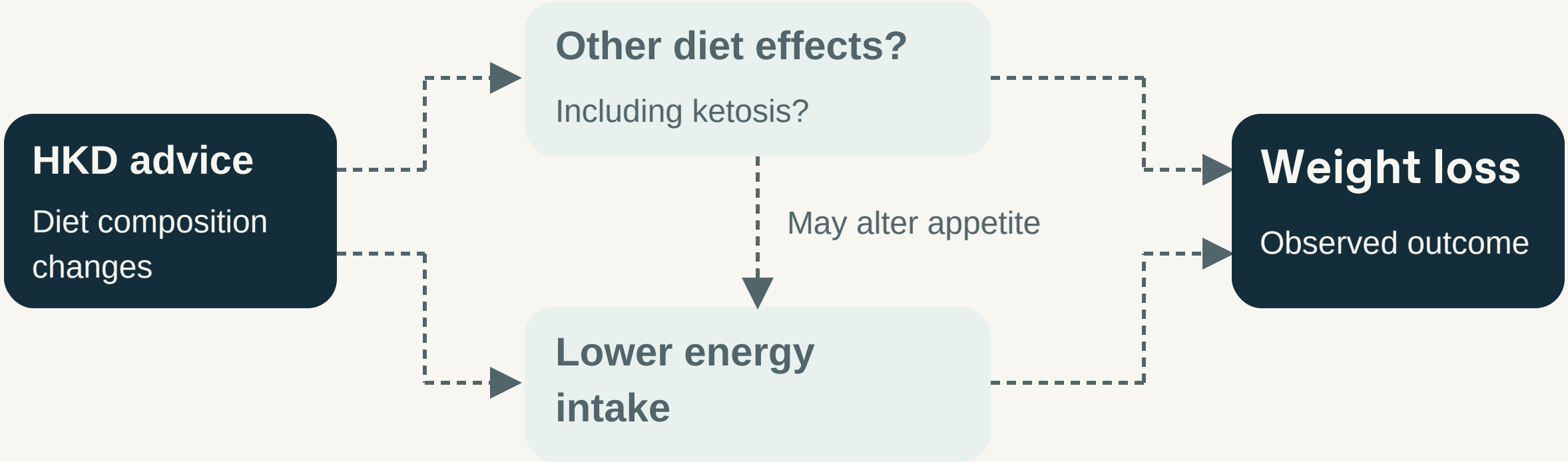
kcal/day

Both groups were advised to restrict energy.

Reported calorie reductions still differed.

Lim et al. 2024 · Table 5 · These are reported changes in intake, not prescribed or absolute daily calories.

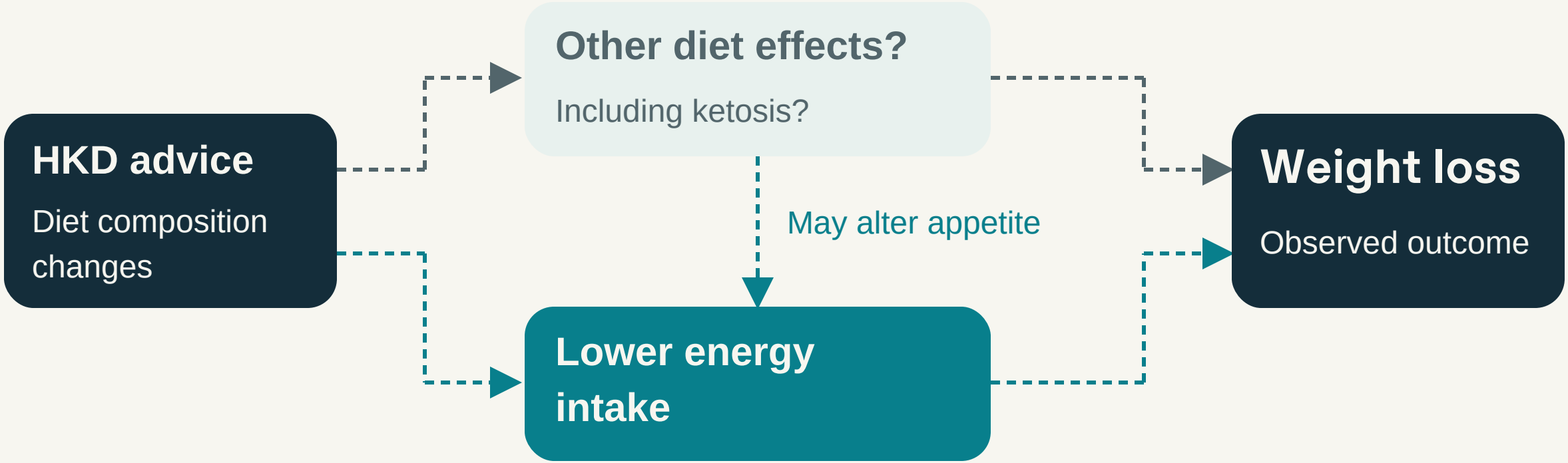
Two routes could explain it.



Possible pathways can operate together; neither was isolated.

Interpretation of Table 5 + Discussion · Dashed arrows are hypotheses, not demonstrated mediation.

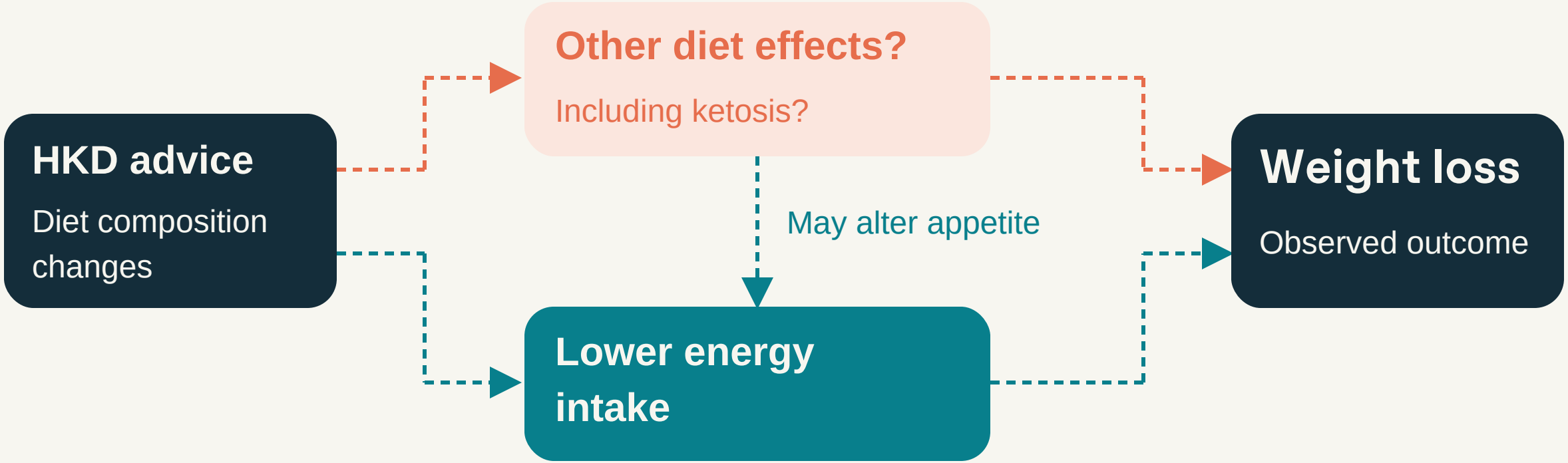
Eating less may be the pathway.



Mediator = a step through which the programme may work.

Interpretation of Table 5 + Discussion · Dashed arrows are hypotheses, not demonstrated mediation.

The programme is not the mechanism.



Not automatically confounding or failed randomisation.

A calorie-independent effect was not isolated.

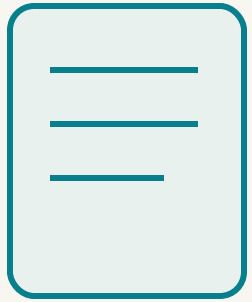
Interpretation of Table 5 + Discussion · Dashed arrows are hypotheses, not demonstrated mediation.

A prescription is not exposure.

THE PRESCRIPTION

≤50 g

net carbohydrate/day



Food diaries



Ketosis?

A diary reports intake.

It does not measure a metabolic state.

Who actually entered ketosis?

Assigned HKD

✓ Known label

Blood or urine ketone checks?

Not reported



Unknown exposure

Who entered ketosis?

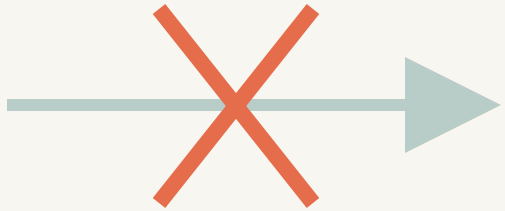
And who stayed in ketosis over time?

Lim et al. 2024 · Discussion · Unknown ketosis does not mean no participant achieved ketosis.

Weight loss cannot verify ketosis.



Weight loss was observed



Does not prove



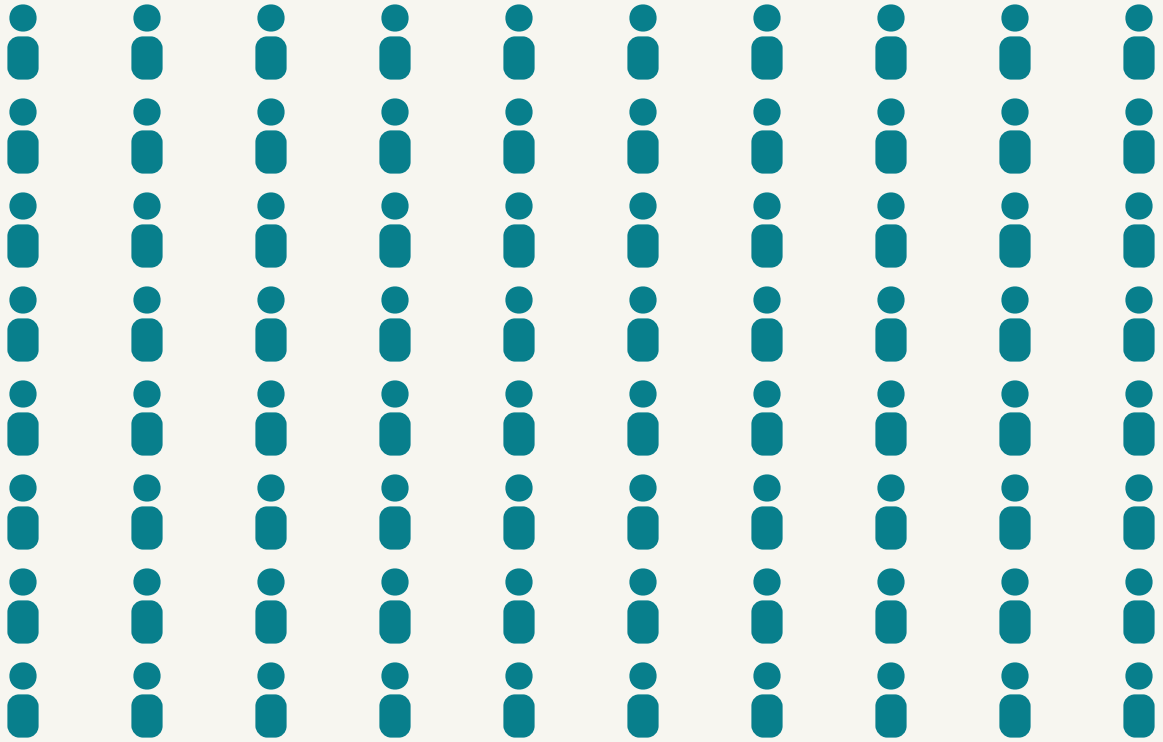
Confirmed ketosis

Calorie reduction can also produce weight loss.

The outcome does not identify a ketosis-specific effect.

Methods 2.3, 2.7 + Discussion · The trial tests assigned advice, not experimentally verified ketosis exposure.

Who stays in the picture?



Observed Same 80-person cohort

80

randomized at baseline

100% in the starting cohort

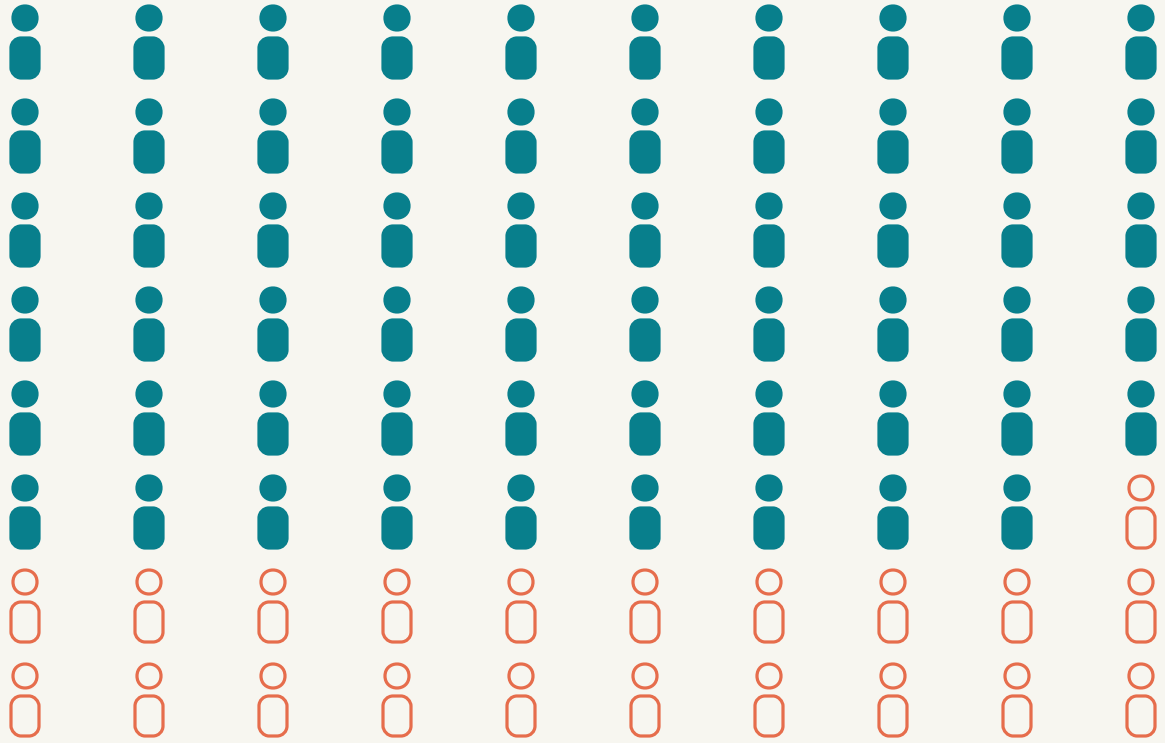
What happens next?

Follow the same cohort.

One figure = one participant

Lim et al. 2024 · Figure 1 + Table 2 · 6-month counts: HKD 29/41; ERD 30/39.

Who stays in the picture?



● Observed ○ Missing at 6 months

59

measured at 6 months

74% retained

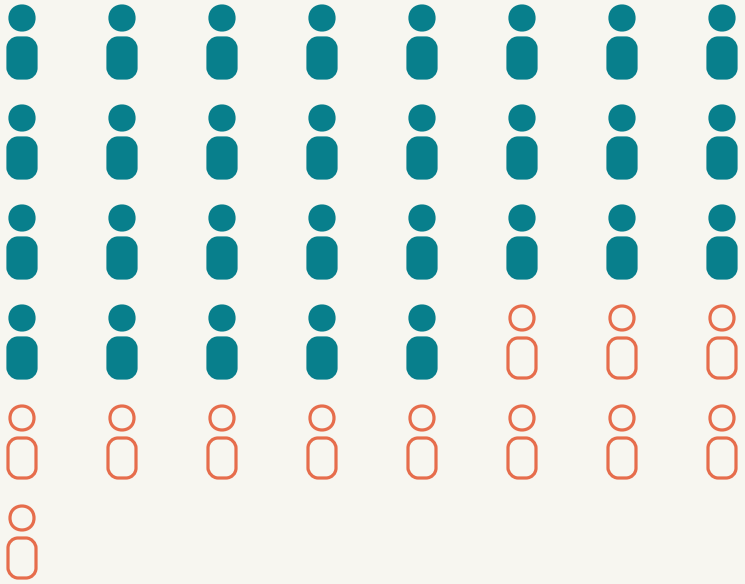
21 outcomes unknown

Bias direction is uncertain.

HKD missing: 12 · ERD: 9

Lim et al. 2024 · Figure 1 + Table 2 · 6-month counts: HKD 29/41; ERD 30/39.

The missing weights could matter.



HKD: 29 measured + 12 missing

THE UNOBSERVED HKD OUTCOMES

12

Missing weights

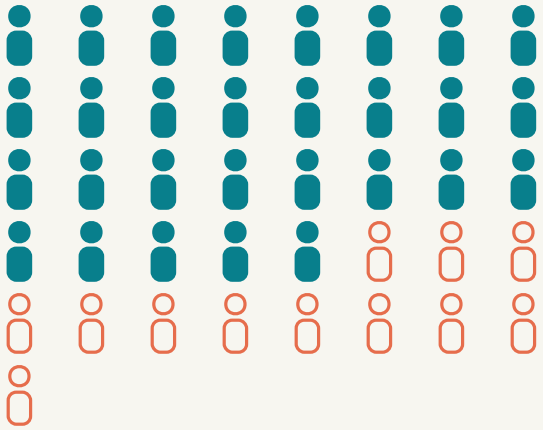
What if they lost less weight?

Their responses are unknown.

The 12 unobserved weights could change the HKD group estimate.

Lim et al. 2024 · Figure 1 + Table 2 · The question is hypothetical; the missing patients' weight changes are unknown.

Selection can change the comparison.



29 measured
12 missing

IF POORER HKD RESPONDERS LEAVE…

The measured HKD subset contains better responders.

Completer-only weight loss could look larger.

Potential attrition bias. Other missingness patterns could shrink the gap.

Randomisation protects the assigned comparison; a subset selected after assignment need not retain that balance.

An ITT label cannot restore weights.

ITT was claimed.

Keeping original group labels does not supply missing weights.

Still needed:

Observed outcomes, or explicit missing-data assumptions.

Handling is unclear in the report.

Diet stopped ≠ follow-up stopped.

Their outcomes still belong to the assignment-group comparison.

Lim et al. 2024 · Methods 2.9 · Continue outcome collection where possible; remaining missingness requires assumptions.

A result is not a full explanation.

WHAT THE DESIGN CAN EVALUATE

A supported dietary programme.

ENERGY	Calorie-independent effects were not isolated.
KETOSIS	Actual metabolic exposure was not verified.
ATTRITION	Missing outcomes may change the comparison.

Lim SL et al. (2024). Nutrients 16:4380. doi:10.3390/nu16244380